부산경남 1차 미니 프로젝트

AI 10반 28조

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개요

Day 1

데이터 전처리 & 탐색

데이터의 분석 및 변수 중요도
딥러닝을 위한 데이터 전처리 및 모델의 비교

V

모델 리뷰

Base/hidden layer/drop out/
early stopping/drop out+early stopping

V

Day 2

데이터 전처리 파이프라인

데이터 전처리 함수

V

기본모델 예측 및 평가

새로운 데이터를 활용하여
기존 모델 예측 및 평가

V

모델 추가 학습 및 미세조정

새로운 데이터 추가학습 + 성능예측

데이터 분석 및 전처리

DATA.INFO()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 19278 entries, 0 to 19277
Data columns (total 25 columns):
 #   Column                             Non-Null Count  Dtype
---  ------                             --------------  -----
 0   Unnamed: 0                         19278 non-null  int64
 1   ID                                 19278 non-null  int64
 2   Gender                             19278 non-null  object
 3   Customer Type                      19278 non-null  object
 4   Age                                18626 non-null  float64
 5   Type of Travel                     19278 non-null  object
 6   Class                              19278 non-null  object
 7   Flight Distance                    19278 non-null  int64
 8   Inflight wifi service              18640 non-null  float64
 9   Departure/Arrival time convenient  19278 non-null  int64
 10  Ease of Online booking             19278 non-null  int64
 11  Gate location                      19278 non-null  int64
 12  Food and drink                     18893 non-null  float64
 13  Online boarding                    19278 non-null  int64
 14  Seat comfort                       18791 non-null  float64
 15  Inflight entertainment             19278 non-null  int64
 16  On-board service                   19278 non-null  int64
 17  Leg room service                   19278 non-null  int64
 18  Baggage handling                   19278 non-null  int64
 19  Checkin service                    19278 non-null  int64
 20  Inflight service                   19278 non-null  int64
 21  Cleanliness                        19278 non-null  int64
 22  Departure Delay in Minutes         19278 non-null  int64
 23  Arrival Delay in Minutes           19224 non-null  float64
 24  Satisfaction                       19278 non-null  object
dtypes: float64(5), int64(15), object(5)
memory usage: 3.7+ MB
```

불필요한 변수

부적합한 데이터 타입

결측치

관측값

불필요한 변수

```python
drop_cols = ['ID','Unnamed: 0']
data.drop(columns=drop_cols, inplace=True)
```

부적합한 데이터 타입

```python
# 문자열 변수 리스트로 선언
cat_cols = ['Gender','Customer Type','Type of Travel','Class']

# 라벨 인코딩
for col in cat_cols:
    encoder = LabelEncoder()
    data[col] = encoder.fit_transform(data[col])
```

결측치

```python
# 수치형 변수 선택
num_cols = data.select_dtypes(include=['number']).columns

# 결측치 → 중앙값
data[num_cols] = data[num_cols].fillna(data[num_cols].median())
```

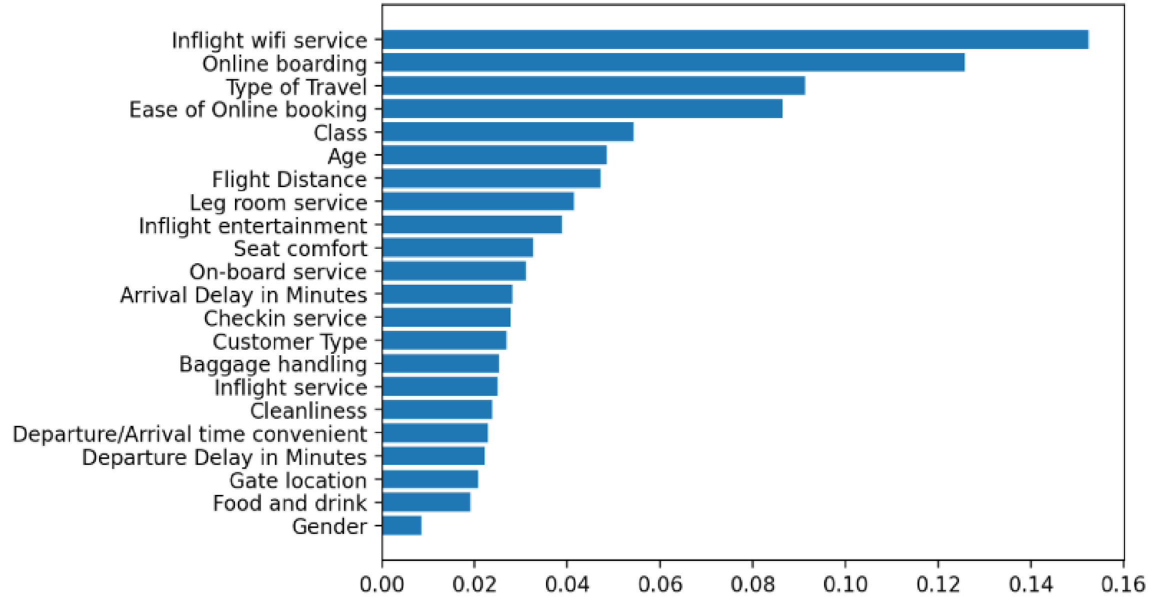
```python
# 범주형 변수 선택
cat_cols = data.select_dtypes(include=['object', 'category']).columns

# 결측치 → 최빈값
data[num_cols] = data[num_cols].fillna(data[num_cols].mode().iloc[0])
```

관측값

```python
target = 'Satisfaction'
data['Satisfaction'] = data['Satisfaction'].map({
                          'Satisfied': 1,
                          'Neutral or Dissatisfied': 0
                       })
```

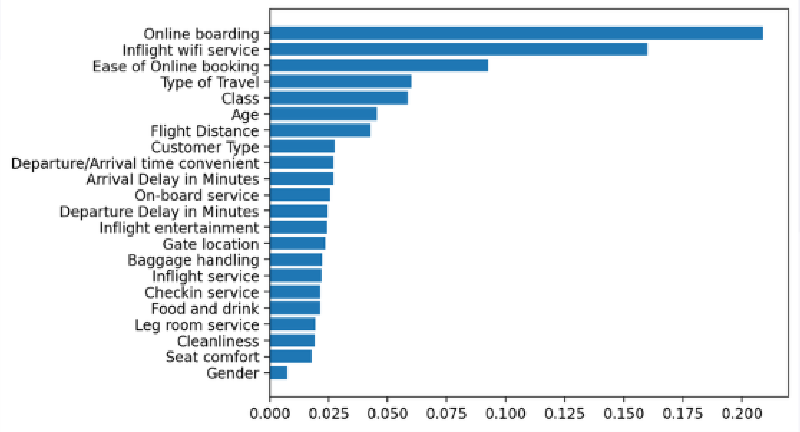
데이터 분석 및 전처리

변수 중요도

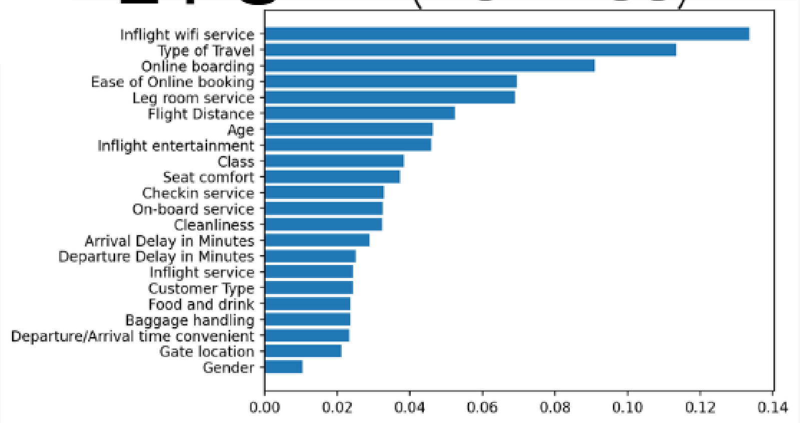


모델: RANDOMFORESTCLASSIFIER

변수 중요도 (AGE <= 35)



변수 중요도 (AGE > 35)



MODEL REVIEW

가변수화

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 19278 entries, 0 to 19277
Data columns (total 79 columns):
                                          non-null  int64
                                          non-null  int64
                                          non-null  float64
 3   Type of Travel                 19278 non-null  int64
 4   Class                          19278 non-null  int64
 5   Flight Distance                19278 non-null  int64
 6   Departure Delay in Minutes     19278 non-null  int64
 7   Arrival Delay in Minutes       19278 non-null  float64
 8   Satisfaction                   19278 non-null  int64
```

```
 10  Inflight wifi service_2              19278 non-null  int64
 11  Inflight wifi service_3              19278 non-null  int64
 12  Inflight wifi service_4              19278 non-null  int64
 13  Inflight wifi service_5              19278 non-null  int64
 14  Departure/Arrival time convenient_1  19278 non-null  int64
 15  Departure/Arrival time convenient_2  19278 non-null  int64
 16  Departure/Arrival time convenient_3  19278 non-null  int64
 17  Departure/Arrival time convenient_4  19278 non-null  int64
 18  Departure/Arrival time convenient_5  19278 non-null  int64
 19  Ease of Online booking_1             19278 non-null  int64
 20  Ease of Online booking_2             19278 non-null  int64
 21  Ease of Online booking_3             19278 non-null  int64
 22  Ease of Online booking_4             19278 non-null  int64
 23  Ease of Online booking_5             19278 non-null  int64
 24  Gate location_1                      19278 non-null  int64
 25  Gate location_2                      19278 non-null  int64
 26  Gate location_3                      19278 non-null  int64
 27  Gate location_4
 28  Gate location_5
```

가변수화 변수 일부

데이터 분할

```python
target = 'Satisfaction'

x = data.drop(target,axis=1)
y = data.loc[:,target]

x_train, x_val, y_train, y_val = train_test_split(x,y,test_size=0.2, random_state=1)
```

78개의 입력변수와
1개의 타겟변수 생성

학습용 데이터 80%   검증용 데이터 20%

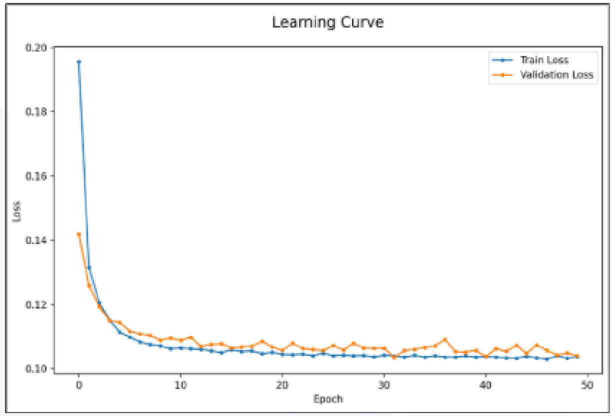
스케일링

```python
scaler = MinMaxScaler()
x_train = scaler.fit_transform(x_train)
x_val = scaler.transform(x_val)
```

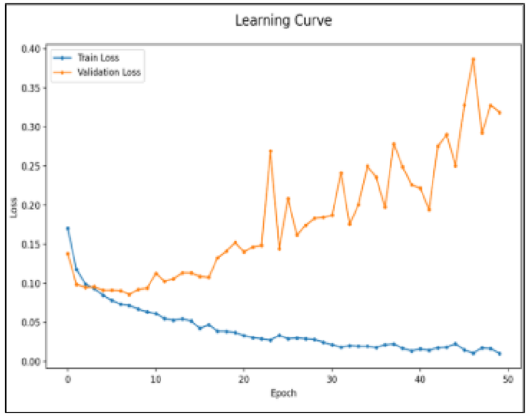
0~1 정규화 작업

MODEL REVIEW

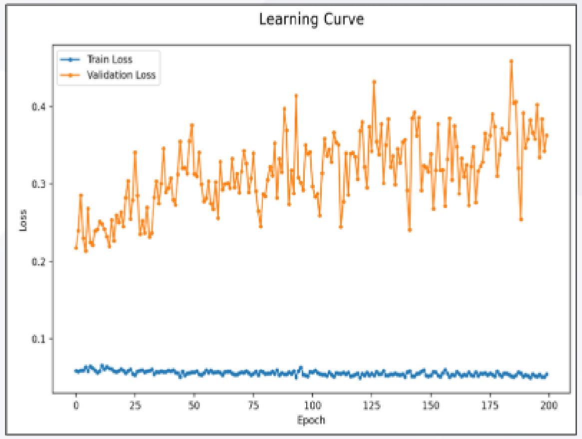
BASE



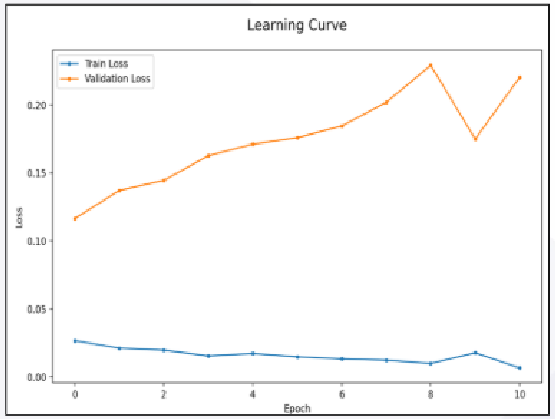
HIDDEN LAYER



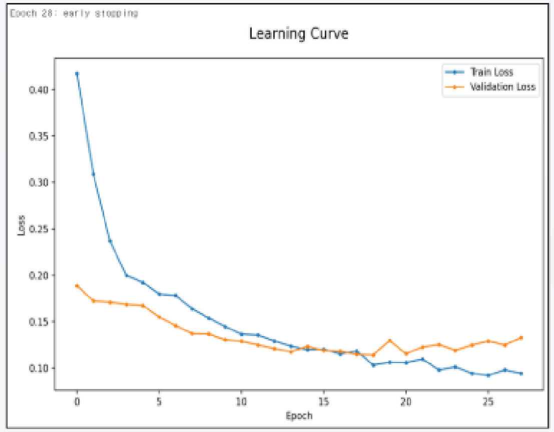
DROP OUT



EARLY STOPPING



DROP OUT + EARLY STOPPING

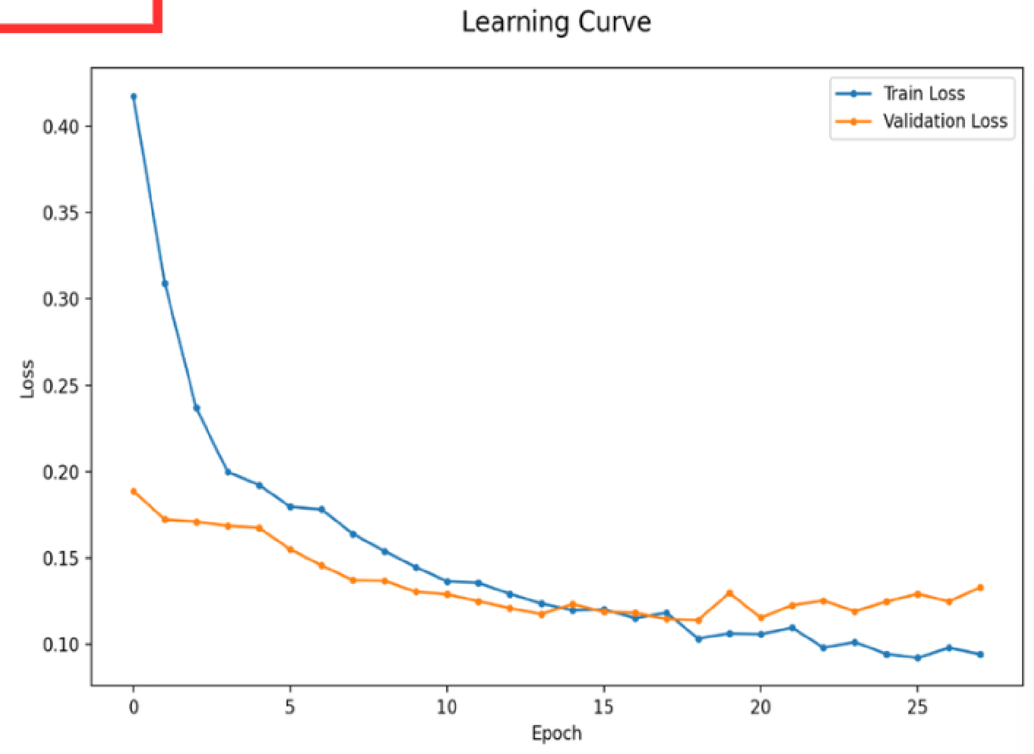


MODEL REVIEW

28조 최적모델

drop out + early stopping

```
Epoch 28: early stopping
```



Dropout(0.3)
learning_rate=0.001
epochs=200

```
121/121 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step
              precision    recall  f1-score   support

           0       0.84      0.82      0.83       358
           1       0.98      0.98      0.98      3498

    accuracy                           0.97      3856
   macro avg       0.91      0.90      0.91      3856
weighted avg       0.97      0.97      0.97      3856
```

데이터 전처리 파이프라인

```python
# 함수 만들기
def build_model_input(data):

  # 1. 결측치 처리
  df = data.copy()

  #수치형 -> 중앙값
  num_cols = df.select_dtypes(include=['number']).columns
  df[num_cols] = df[num_cols].fillna(df[num_cols].median())

  #범주형 -> 최빈값
  cat_cols = df.select_dtypes(include=['object','category']).columns
  df[cat_cols] = df[cat_cols].fillna(df[cat_cols].mode().iloc[0])

  # 2. 값 변경
  target = 'Satisfaction'
  df['Satisfaction'] = df['Satisfaction'].map({
                          'Satisfied': 1,
                            'Neutral or Dissatisfied': 0
                          })

  # 3. 변수 제거
  drop_cols = ['ID','Unnamed: 0']
  df.drop(columns=drop_cols, inplace=True)


  # 4. 라벨 인코딩
  cat_cols = ['Gender','Customer Type','Type of Travel','Class']
  for col in cat_cols:
    encoder = LabelEncoder()
    df[col] = encoder.fit_transform(df[col])
```

```python
# 5. 가변수화
# 설문 범주 정의
categories = [0, 1, 2, 3, 4, 5]

# 가변수화 대상 변수
dumm_cols = ['Inflight wifi service', 'Departure/Arrival time convenient',
             'Ease of Online booking', 'Gate location', 'Food and drink',
              'Online boarding', 'Seat comfort', 'Inflight entertainment',
              'On-board service',  'Leg room service', 'Baggage handling',
             'Checkin service', 'Inflight service', 'Cleanliness']

# Category 형 변수로 변환
for col in dumm_cols:
  df[col] = pd.Categorical(df[col], categories=categories)

# 가변수화
df = pd.get_dummies(data=df, columns=dumm_cols, drop_first=True, dtype=int

# 6. x, y 분리
x=df.drop(columns='Satisfaction')
y=df.loc[:,'Satisfaction']

# 7. 스케일링
x = scaler.transform(x)

# 반환
return x, y
```

기본모델 예측 및 평가

기본 모델

```
y_pred = base_model.predict(x_val)
y_pred = np.where(y_pred>=0.5,1,0)

print(classification_report(y_val,y_pred))
```

```
11/11 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step
              precision    recall  f1-score   support

           0       0.98      0.80      0.88       191
           1       0.80      0.98      0.88       159

    accuracy                           0.88       350
   macro avg       0.89      0.89      0.88       350
weighted avg       0.90      0.88      0.88       350
```

미세조정 모델

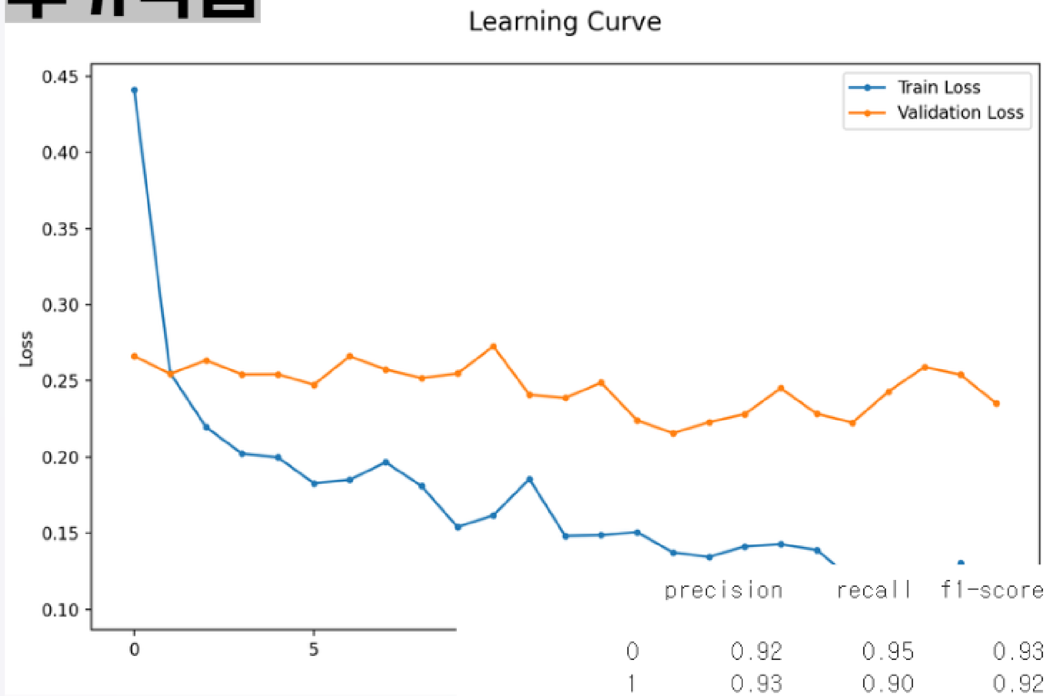
```
y_pred = model.predict(x_val)
y_pred = np.where(y_pred >= 0.5,1,0)
print(classification_report(y_val,y_pred))
```

```
11/11 ━━━━━━━━━━━━━━━━━━━━ 1s 30ms/step
              precision    recall  f1-score   support

           0       0.95      0.93      0.94       191
           1       0.92      0.94      0.93       159

    accuracy                           0.94       350
   macro avg       0.94      0.94      0.94       350
weighted avg       0.94      0.94      0.94       350
```

모델 추가 학습 및 미세조정

```
model = base_model
hist = base_model.fit(x_train,
                      y_train,
                      epochs=30,
                      callbacks=[es],
                      validation_split=0.2
                      ).history
```

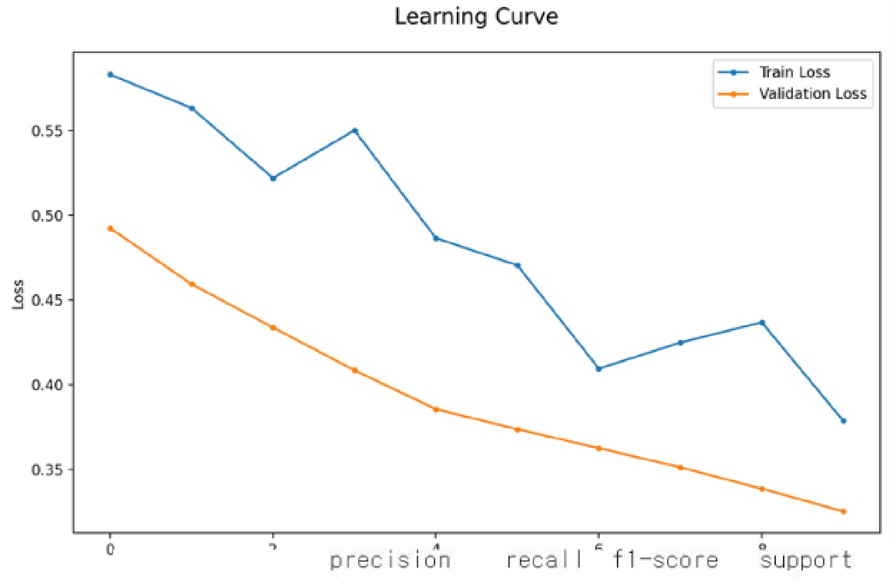
추가학습



	precision	recall	f1-score	support
0	0.92	0.95	0.93	191
1	0.93	0.90	0.92	159
accuracy			0.93	350
macro avg	0.93	0.92	0.92	350
weighted avg	0.93	0.93	0.93	350

```
for layer in base_model.layers[:-3]:
  layer.trainable=False
#모델 선언
model = Sequential([
    base_model,
    Dense(8,activation='relu'),
    Dense(1,activation='sigmoid')
])

model.compile(optimizer=Adam(learning_rate=0.001),
              loss='binary_crossentropy')
hist = model.fit(x_train,
                 y_train,
                 epochs=100,
                 callbacks=[es
                 validation_sp
```

미세조정



	precision	recall	f1-score	support
0	0.95	0.93	0.94	191
1	0.92	0.94	0.93	159
accuracy			0.94	350
macro avg	0.94	0.94	0.94	350
weighted avg	0.94	0.94	0.94	350